



EUROQOL

EuroQol Projects Submission Portal (Current)

2582-SG 1.3 Seed grant

EQ-Graph



wrQGBmAn

Submission details

My submission

Full Title

| Creating a EuroQol Research Knowledge Graph

EuroQol Seed Grants are available to support innovative research that may result in providing a catalyst to, and a starting point for, launching new projects or facilitating a change in the direction of the research.

Seed Grants are not solely intended for research and can also give an opportunity for collaboration; to make new networks and extend existing ones.

Through a Seed Grant, short-term, exploratory research projects can be supported. The seed funds are provided to stimulate innovative use of EQ-5D or of EQ-5D data. The results will determine the potential and long-term feasibility of the research or inform the design and execution of a larger scale study. A seed grant allows researchers to explore new research initiatives, that may be high-risk projects, to start working on their research project.

Examples include:

- Applications leveraging further value from existing data sets made available through Routine Outcome Monitoring
- Development and validation of risk-prediction models, diagnostic tools, or decision support models
- Studies addressing the nature of disparities in care and outcomes
- Research done in strategic framework for value-based health care research and/or implementation
- Patient profiles in health care, patient platforms, care pathways
- Methodological or applied research in the field of psychometrics, health economics, classification theory, mapping, instrument development, or information processing and presentation.

Abstract.

Please provide a brief summary of your proposal. Provide enough information to enable the reader to understand what will be done, and how. Please note that the abstract will be posted on the EuroQol website for successful applications.

EuroQol has funded over 900 research projects, producing a large body of publications on EQ-5D and related instruments. This literature contains detailed evidence on how these instruments perform across populations,

countries, clinical areas, and methods. However, this information is distributed across individual papers and is not easily searchable or aggregable at the portfolio level.

We propose to build a research knowledge graph of EuroQol's funded projects and their associated publications. Using large language models (LLMs), we will extract structured data from full-text papers: study designs, sample characteristics, instrument versions, methods, countries, and key findings. The extracted data will be stored in a graph database and made accessible through a web application that supports structured search, visual summaries, and natural-language queries.

The project will proceed in three stages: (1) assembling the corpus from EuroQol's project database and identifying associated publications; (2) developing and iteratively refining the extraction pipeline; (3) building the web application. A preliminary demonstration, based on 944 project abstracts, produced a graph of over 2,000 nodes and 8,000 edges. The seed grant will go further by extracting from full-text publications and providing an interactive interface.

If the approach proves useful, the knowledge graph can be expanded to cover the broader EQ-5D literature (over 17,000 publications in PubMed). The seed grant produces the working system and the evidence needed to assess whether such an expansion is warranted.

Budget requested from EuroQol (euro)	€28,404.00
--------------------------------------	------------

Are you receiving co-funding or in-kind support from another organisation?

Relevant Working Group(s) ☒ Others

EuroQol has launched a **SAVED** archive intended to make data available for secondary use, where datasets are considered useful or valuable. For more information, please visit EuroQol's [SAVED webpage](#).

All submissions will be reviewed to see if they are suitable to receive SAVED status. To assist us with this review, please indicate in which areas of research this dataset could be re-used for.

N/A

Research strategy

Please provide a brief description of your organisation and its main research area and achievements.

Shoulders (<https://should.rs>) is an independent research and development group building AI tools for researchers. Its core product is an open-source AI workspace. It integrates document editing, reference management, and AI assistance into a single environment for researchers.

Shoulders' broader aim is to make AI practically useful in research workflows. Current work includes structured information extraction from research literature, AI-assisted literature search, and interfaces that make knowledge bases accessible to non-technical users.

Shoulders was founded by Paul Schneider, a research software engineer with a background in health economics, clinical medicine, and epidemiology. Paul has previously collaborated on several EuroQol-funded projects related to valuation methodology, but is not a member of the EuroQol Group. This proposal does not build on that earlier work. It is a new line of research, applying AI-based knowledge extraction and graph infrastructure to EuroQol's research portfolio.

Details of your research plan

(1) Background to the Problem

Over the past decades, EuroQol has funded several hundred research projects. The instruments developed and maintained by EuroQol are among the most widely used patient-reported outcome measures in health research. Over 17,000 publications indexed in PubMed reference EQ-5D. These span valuation, psychometrics, clinical application, population health surveys, and health economic evaluation across a wide range of countries and clinical areas.

This literature contains detailed information about how EQ-5D instruments perform, what they measure, and how they are used in practice. However, that information is distributed across thousands of individual papers, each reporting on specific study designs, populations, methods, and findings.

Identifying which populations have been studied with a particular instrument version, or which methods have been applied in a given clinical area, typically requires manual literature review. This is feasible for narrow questions, but becomes increasingly difficult as the evidence base grows.

Research knowledge graphs provide one approach to this problem. A knowledge graph stores information as entities (e.g. studies, instruments, populations, countries) and the relationships between them. For example, a valuation study would be linked to the instrument it valued (EQ-5D-5L), the country it was conducted in (Germany), and the method it used (composite TTO). A clinical population (e.g. stroke patients) would be linked to all studies that recruited from it.

Because the information is stored as structured entities and relationships rather than as free text, it can be queried. A researcher could ask, for instance, "which countries have EQ-5D-5L value sets?" or "which populations have been studied using Rasch analysis?" and immediately get a complete answer from across the entire corpus, without having to search through individual papers.

Knowledge graphs are established infrastructure in several research domains. Notable examples include:

- **OpenAlex** : indexes over 300 million scholarly works as a knowledge graph of publications, authors, institutions, and concepts (Priem et al. 2022).
- **Cochrane Linked Data**: organises clinical trial evidence using a PICO ontology to connect interventions, outcomes, and populations (Mavergames et al. 2019).
- **Hetionet**: integrates 47,000 biomedical entities from 29 data sources; constructed during a single PhD project (Himmelstein et al. 2017).
- **PrimeKG**: contains 129,000 nodes supporting precision medicine research; built by a three-person team (Chandak et al. 2023).
- **HERCULES**: links researchers, projects, and publications within a national university system for research management (Chaves-Fraga et al. 2022).

Until recently, building a knowledge graph from research literature required extensive manual coding. Trained researchers had to read each paper and extract structured data by hand — a process that, for a corpus of several hundred publications, would take months of full-time work and require ongoing effort to maintain as new studies are published.

Recent advances in large language models (LLMs) have changed this. Current models can read full-text papers, identify relevant entities and relationships, and output structured data, which means that much of the manual reading and coding work can now be automated.

A study in the *Annals of Internal Medicine* found that LLM-assisted extraction in systematic reviews achieved 91.0% accuracy, compared to 89.0% for human-only extraction (Gartlehner et al. 2025). Several studies have also demonstrated the feasibility of constructing biomedical knowledge graphs directly from research text using LLMs (Yang et al. 2024; Arsenyan et al. 2024). These results suggest that automated extraction can now reliably populate a knowledge graph from a corpus of several hundred publications. The researcher still reviews and validates what the

pipeline extracts, and decides how the schema should evolve, but the manual reading and coding is handled by the model.

LLMs also address a second barrier. Traditionally, querying a knowledge graph required writing queries in specialised languages such as SPARQL or Cypher — a skill most health researchers do not have. Current LLMs can translate plain-English questions into graph queries and summarise the results, which means that researchers can interrogate the graph directly without programming expertise.

(2) Aim of the Research

We propose to build a research knowledge graph of all of EuroQol's funded projects and their associated publications. For each funded project, the graph will contain related publications and extract metadata (authors, affiliations, year, journal) as well as detailed structured data from the full text: study designs, sample characteristics, instruments and versions used, methods applied, countries, and key findings.

In a second step, we will make the graph accessible through a web application supporting structured search, visual summaries, and natural-language queries. The seed grant produces a working system for the funded portfolio. The results will inform whether and how the approach could be extended to the broader EQ-5D literature.

(3) Proposed Methods

3.1 Corpus definition and assembly

The corpus will be assembled from EuroQol's publicly available project database, which currently lists 944 funded projects. For each project, we will identify associated publications using EuroQol's internal grant records and funding acknowledgement statements in publications. If more detailed project documentation is available from the EuroQol Group (e.g. project reports or internal records), these could be used for richer extraction.

Publication metadata (titles, authors, abstracts, DOIs) will be collected via the OpenAlex and Semantic Scholar APIs. Full-text PDFs will be retrieved through institutional access. Matching publications to funded projects, for instance by identifying funding acknowledgement statements, will itself be partly automated using LLMs.

The EuroQol-funded publications provide a solid foundation for the knowledge graph. If the approach proves useful, the graph can be expanded to include the broader body of EQ-5D research, including studies not directly funded by EuroQol.

3.2 Schema development

Before extracting data, we need to decide what to extract. The schema specifies which types of information the graph will contain, for instance, whether we distinguish between instrument versions, whether we record sample sizes, or whether we capture the statistical methods used. The initial schema will be informed by EuroQol's own instrument taxonomy and by existing frameworks in health outcomes research.

Core entity types are likely to include: Study, Instrument (version-specific, e.g. EQ-5D-3L, EQ-5D-5L, EQ-5D-Y, EQ-HWB), Population, Method, Country, Author, Institution, Working Group, Value Set, and Outcome.

Core relationships will capture which instrument a study used, which population it studied, which methods it applied, which country it was conducted in, which value set it produced, and which EuroQol project funded it.

The final schema will emerge from the data itself: as the extraction pipeline encounters information not captured by the initial categories, the schema is expanded (see Section 3.4). However, the initial version described above provides the starting structure.

3.3 LLM extraction pipeline

The extraction pipeline will read each paper and extract structured data against the current schema. For each paper, the extracted information will include:

- Study design (e.g. cross-sectional survey, RCT, cohort study, valuation study)
- Sample characteristics (size, age range, sex distribution, clinical condition, severity, recruitment setting)
- Instrument versions and administration modes (self-report, proxy, interviewer-administered)
- Statistical methods and models (e.g. composite TTO, DCE with conditional logit, Rasch analysis, IRT)
- Comparator instruments used alongside EQ-5D
- Key findings (value set coefficients, measurement properties, population norms)

To assess accuracy, a research assistant will independently read and code approximately 50 papers by hand. The pipeline's output will then be compared against this human-coded reference set, and we will report the proportion of correctly extracted items for each type of information (Yang et al. 2024; Arsenyan et al. 2024).

3.4 Iterative schema refinement

It is not possible to anticipate all relevant categories before extraction begins. The literature may contain study types, methods, or population characteristics that were not included in the initial schema. Rather than specifying everything upfront, the schema is refined iteratively as the extraction proceeds.

In practice, this works as follows. The pipeline extracts data from a batch of papers and flags cases where the information does not fit the existing categories. A researcher reviews these cases and decides whether the schema should be expanded. Most of the work (reading papers, identifying entities, proposing new categories) is done by the LLM. The researcher provides oversight, deciding what belongs in the schema and what does not.

3.5 Graph database and access layer

The extracted data will be stored in a graph database (Neo4j, a widely used open-source database designed for storing and querying connected data). A web application will make the graph accessible to EuroQol members and will provide three modes of access.

1. Visual summaries: The application will provide pre-built overviews of the funded portfolio, including geographic coverage, instrument usage over time, and distributions of study types and methods.

2. Structured search: Researchers will be able to filter the graph by instrument, country, study type, method, population, and year, and export the results to Excel or CSV for further analysis.

3. Natural-language queries: Researchers will be able to ask questions in plain English (e.g. "Which funded studies have tested EQ-5D-5L measurement properties in mental health populations?"). The application translates the question into a graph query and returns results with references to the source papers.

3.6 Preliminary illustration

To illustrate the proposed approach, we prepared a preliminary demonstration using publicly available project summaries from EuroQol's funded projects database.

We extracted entities and relationships from 944 project abstracts, producing a graph of 2,116 nodes and 8,213 edges covering projects, researchers, instruments, countries, and methods. This prototype uses only abstract-level data. Summary results are available at: <https://shoulde.rs/eq-graph>

The seed grant would go substantially further: extracting detailed structured data from full-text publications and making the structured data accessible through an interactive, AI-augmented web application.

(4) What will the EuroQol Group learn/gain by funding this research proposal

This seed grant will produce a knowledge graph containing structured data from EuroQol's funded projects and their publications: which instruments were used, in which populations, with which methods, in which countries, and what the key findings were. For the first time, this information will be searchable across the entire funded portfolio rather than only accessible by reading individual papers.

The graph could support several types of use:

- Locating studies matching specific criteria across the funded portfolio (e.g. all valuation studies using composite TTO in European countries)
- Providing visual overviews of coverage by instrument, country, population, and method
- Answering ad hoc questions about the funded research base in plain English
- Exploring methodological trends, geographic patterns, and connections between funded projects
- Linking studies in the graph to datasets in the SAVED archive

Beyond search and exploration, the knowledge graph makes certain types of research substantially easier or possible for the first time. Systematic reviews of EQ-5D studies, which currently require weeks or months of manual screening, could be supported by pre-structured evidence from the graph. Rapid scoping reviews (e.g. "what is known about EQ-5D-Y in low-income countries?") could be answered in minutes rather than days. The graph could also be used to identify gaps in the evidence base, such as populations or countries where a particular instrument version has not yet been studied.

This seed grant focuses on EuroQol's funded portfolio. The knowledge graph itself can be expanded. If the approach proves useful, a follow-on project could extend the graph to the broader EQ-5D literature (over 17,000 publications indexed in PubMed), automate updating as new papers are published, and develop deeper query and synthesis capabilities.

Please list the deliverables of your research and describe their value to EuroQol

1. Populated knowledge graph of EuroQol's funded projects
2. Web application with AI-assisted search and visualisation (12 months hosting)
3. Open-source code repository
4. Plenary presentation with live demonstration
5. Final report

How does your proposal align with EuroQol's strategic plan?

This seed proposal brings AI and data engineering capability into EuroQol's research programme. The knowledge graph would complement existing EuroQol resources (the publications database, SAVED archive) by adding structured, searchable data from within individual papers. It could help Working Groups understand what has been

studied with a given instrument or in a given population, help researchers locate prior work when preparing proposals, and support research priority setting across the Group.

Please outline the plans for disseminating or implementing the results of the research

- Plenary/Academy meeting presentation and live demonstration (2027)
- Open-source code in a public repository
- Web application accessible to EuroQol members and the broader research community

What support would you like to receive from EuroQol for this project, besides financial support?

Access to EuroQol's internal project records are not required but would improve the quality of the knowledge graph. In particular, we would benefit from any of the below.

- The funded projects database, including any fields not publicly available (e.g. links to publications, project reports, timelines, final reports)
- Working Group members willing to review the schema or provide feedback on the web application
- Access to the SAVED archive metadata, to enable linking graph entries to archived datasets

Proposed start date 2026-07-01

Duration of your project (in months) 6

Proposed completion date 2026-12-31
(performance period end date)

Please note, for Seed Grant applications the duration of the project must not exceed 12 months

Please describe the timelines of the planned research. If you prefer to upload this as an attachment, please indicate that in the field

Please see attached Gantt chart.

Does your research require the development of a modified or new version of an EQ instrument (changed wording, bolt ons, new language version, new mode of administration)? ✓ no, all required versions are available

How are stakeholders involved in your project?

The main stakeholders are EuroQol members. We plan to present the project and demonstrate the web application at the 2027 Plenary or Academy Meeting, which will be the main opportunity for feedback. If Working Group members are willing to review the schema or test the application during development, we would welcome their input, but this is not a requirement for completing the project.

Data security and confidentiality

This project does not collect personal data. All data used in the knowledge graph are extracted from published research papers and publicly available project records. If internal EuroQol documentation is shared with the project team, it will be stored securely and used only for the purposes described in this proposal. The web application and graph database will be hosted on encrypted cloud infrastructure with access controls.

Registration to database / repository

N/A

Ethics approval status not required

Did you use artificial intelligence (AI) in your proposal? ☒ Yes

Please explain in detail how AI was used in your proposal

AI (Claude Opus 4.6) was used to assist with revising the text of the proposal. All content was drafted, reviewed and edited by the applicants.

For funded proposals, a grant offer letter will be sent out via DocuSign for signature. Please provide the name, full address, and email address of the individual to whom the grant offer letter should be addressed and sent for signature. This person must have the legal authority to sign the grant offer letter (e.g., a grants or legal department representative in a university, or yourself if you are applying independently). Before submitting the application, please confirm with the relevant parties that the listed individual is the correct authorised signatory. Additional signatories may be added only if required.

Paul Schneider
Gewerkenstr 8
44805 Bochum
Germany
paul@shoulde.rs

If you would like us to send a copy (for viewing only) of the grant offer letter to any additional contacts (before and after signature), please provide the name(s) and email address(es) of the person(s) who should receive the copy.

1	N/A
---	-----

Other comments

A preliminary demonstration of an EQ knowledge graph from 944 EuroQol project abstracts (cotaining 2,116 nodes and 8,213 edges) is available at <https://shoulde.rs/eq-graph>.

Budget specification (in Euro)

XLSX

[EQ-Graph_Budget.xlsx \(11 KiB download\)](#)

You can upload the CVs of project team members here, all combined in a single file (.doc or .pdf)



[CVs.pdf \(142 KiB download\)](#)

Project team

Name and Surname
Paul Schneider

Email

paul@shoulde.rs

Role

Principal Investigator (PI)

Is this a EuroQol member?

no

To provide a brief CV here, please include i. current position, ii. relevant publications, and iii. relevant grants received**# I Current Position**

- Founder of Shoulders, Bochum, Germany
- AI for Research Fellow, Foresight Institute, Sand Francisco, USA

II Relevant publications

Schneider P, Schramm A. The wisdom of deliberating AI crowds: Does deliberation improve LLM-based forecasting? arXiv preprint arXiv:2512.22625. 2025.

Schneider PP, van Hout B, Heisen M, Brazier J, Devlin N. The online elicitation of personal utility functions (OPUF) tool: a new method for valuing health states. Wellcome Open Res. 2022;7:14.

Schneider P, Devlin N, van Hout B, Brazier J. Exploring health preference heterogeneity in the UK: Using the online elicitation of personal utility functions approach to construct EQ-5D-5L value functions on societal, group and individual level. Health Econ. 2024;33(5):894–910.

Schneider P, Blankart K, Brazier J, van Hout B, Devlin N. Using the online elicitation of personal utility functions approach to derive a patient-based 5-level version of EQ-5D value set: a study in 122 patients with rheumatic diseases from Germany. Value Health. 2024;27(3):376–382.

Schneider P, Ludwig K, Marten O, McDool E, Peasgood T, Devlin N, et al. Using the OPUF approach to create a value set for the EQ-HWB-S: An exploratory feasibility study. Wellcome Open Res. 2024;9.

Hitch J, Kumar G, Schneider P, Devlin N, Shah K, Mott D. The influence of perspective on the valuation of the EQ-5D-Y-3L: A comparison using the OPUF tool and a discrete choice experiment. Pharmacoeconomics. 2026;44(1):61–72.

Moolla A, Schneider P, Hofman K, Goldstein SJ, Thsehla E, Dixon S. Derivation of a South African tariff for the EQ-5D-5L using a personal utility function approach. Int J Technol Assess Health Care. 2025:1–23.

Schneider PP. Social tariffs and democratic choice—Do population-based health state values reflect the will of the people? Health Econ. 2021;30(1):104–112.

McNamara S, Schneider PP, Love-Koh J, Doran T, Gutacker N. Quality-adjusted life expectancy norms for the English population. Value Health. 2023;26(2):163–169.

Smith RA, Schneider PP. Making health economic models Shiny: A tutorial. Wellcome Open Res. 2020;5:69.

III Relevant Grants Received

- Foresight Institute AI for Research Grant. Research Fellow
- Impact of switching value sets across population groups. EQ-Project .RA-2192 Co-I.
- Exploring bolt-on valuation with OPUF. EQ-Project 1807-RA. Co-PI;
- EQ-HWB OPUF Valuation Study. EQ-Project 1508-RA. Co-PI;
- Using OPUF to explore issues in the EQ-5D-Y valuation. EQ-Project 432-RA. Co-I.
- Better than dead? – this or that? Testing how framing impacts viewing a health state as worse than dead. EQ Project 404-RA. Co-PI;
- Using the Online Personal Utility Functions (OPUF) Tool to Explore Issues in the Valuation of EQ-5D-Y. EQ-Project 432-RA. Co-I;
- Estimating quality-adjusted life expectancy for England (DeQoL-LIFE2). EQ Project 123-2020RA-2. Co-I;
- Decomposing the socioeconomic gradient in health-related quality of life over the life course (DeQoL-LIFE). EQ Project 123-2020RA. Co-I;
- Conceptual challenges in the valuation of health in children. EQ Project 117-RA. Co-I;
- Health state utility rescaling and interpersonal comparisons. EQ Project 20190940. co-PI;
- Wellcome Trust funded PhD scholarship within the DTC in Public Health, Economics and Decision Science. The University of Sheffield;

Name and Surname

Sofia Fabishevskaya

Email

sofia@shoulde.rs

Role

Co-Investigator (co-I)

Is this a EuroQol member?

no

To provide a brief CV here, please include i. current position, ii. relevant publications, and iii. relevant grants received

See CV attached

Name and Surname

Kazik Pogoda

Email

kazik@xemantic.com

Role

Advisor (A)

Is this a EuroQol member?

no

To provide a brief CV here, please include i. current position, ii. relevant publications, and iii. relevant grants received

See CV attached

Log in to euroqol-proposals.grantplatform.com to see complete submission attachments.

PDF	
Please upload any other attachment (Excel or PDF) that might be relevant for your submission	
Title of appendix	
Gantt	
Gantt.pdf	97 KiB